

(56)

References Cited**U.S. PATENT DOCUMENTS**

10,505,459	B2	12/2019	Murakami et al.	
11,011,936	B2	5/2021	Costinett et al.	
11,575,315	B2	2/2023	Fu et al.	
2014/0355319	A1 *	12/2014	Cohen	H02M 7/217 363/126
2015/0009734	A1	1/2015	Stahl et al.	
2015/0263605	A1 *	9/2015	Alam	H02M 3/33546 363/21.02
2015/0303813	A1 *	10/2015	Cheng	H02M 1/4258 363/21.02
2017/0290114	A1	10/2017	Glaser et al.	
2017/0338748	A1	11/2017	Liang et al.	
2018/0278177	A1	9/2018	Glovinsky et al.	
2022/0021297	A1 *	1/2022	Xu	H02M 1/4233
2022/0069722	A1	3/2022	Pan	
2022/0247303	A1	8/2022	Jin et al.	
2022/0360196	A1	11/2022	Ilic et al.	
2023/0006540	A1	1/2023	Wang et al.	
2024/0162831	A1	5/2024	Mukherjee et al.	

FOREIGN PATENT DOCUMENTS

CN	109921646	A	6/2019
CN	110089020	B	4/2021
CN	111654196	B	9/2021
CN	112803746	B	4/2022
TW	201633282	A	9/2016
TW	202211604	A	3/2022
TW	202211605	A	3/2022

OTHER PUBLICATIONS

In-Hwan Oh Ed—Anonymous: “A soft-switching synchronous buck converter for Zero Voltage Switching (ZVS) in light and full load conditions” Applied Power Electronics Conference and Exposition, 2008, APEC 2008. Twenty-Third Annual IEEE, Piscataway, NJ, USA, Feb. 24, 2008, pp. 1460-1464; pp. 753, Figs. 1, 3.

Lin, Bor-Ren et al: “Resonant converter with flying capacitors—Analysis and Experiments”, 2014 9th IEEE Conference On Industrial Electronics and Applications, IEEE, Jun. 9, 2014, pp. 161-166, Fig. 1.

Kai Dong et al: “A GaN Totem-Pole PFC Converter with Zero-Voltage-Switching PWM Control”, 2021 IEEE Applied Power Electronics Conference and Exposition (APEC), IEEE, Jun. 14, 2021, pp. 247-251, abstract, Figs. 2(a), 2(b).

N.Z. Yahaya et al: “Design and Simulation of an Improved Soft-Switched Synchronous Buck Converter”, Modelling&Simulation, 2009. AMS '09. Third Asia International Conference On, IEEE, Piscataway, NJ, USA, May 25, 2009, pp. 751-756, pp. 1460-1461; Fig. 1.

Search Report dated Nov. 29, 2023 of the corresponding related European patent application No. 23179269.8.

Search Report dated Nov. 9, 2023 of the corresponding European patent application No. 23177904.2.

Office Action dated Mar. 4, 2024 of the corresponding Taiwan patent application No. 112105302.

Office Action dated May 5, 2025 of the relating U.S. Appl. No. 18/331,383.

* cited by examiner

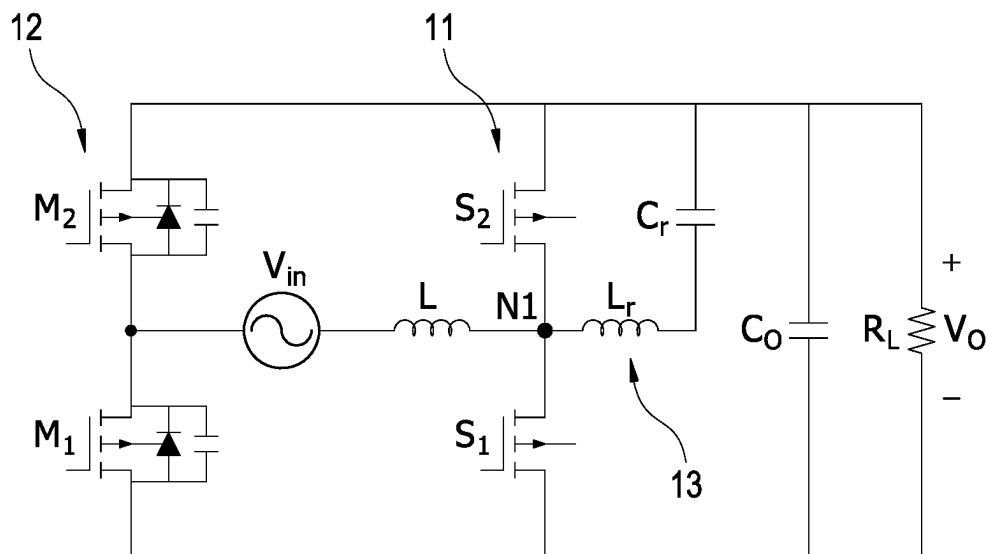


FIG.1A

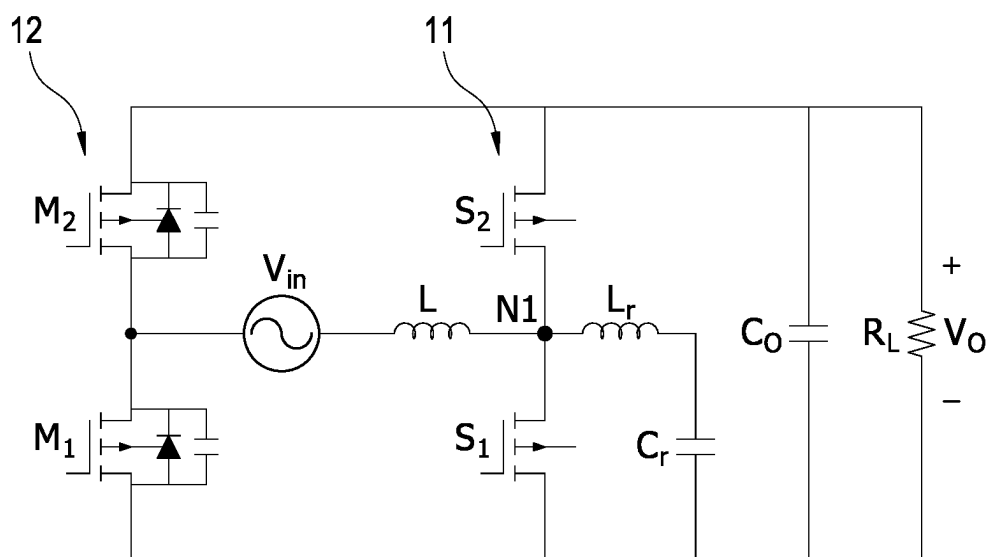


FIG.1B

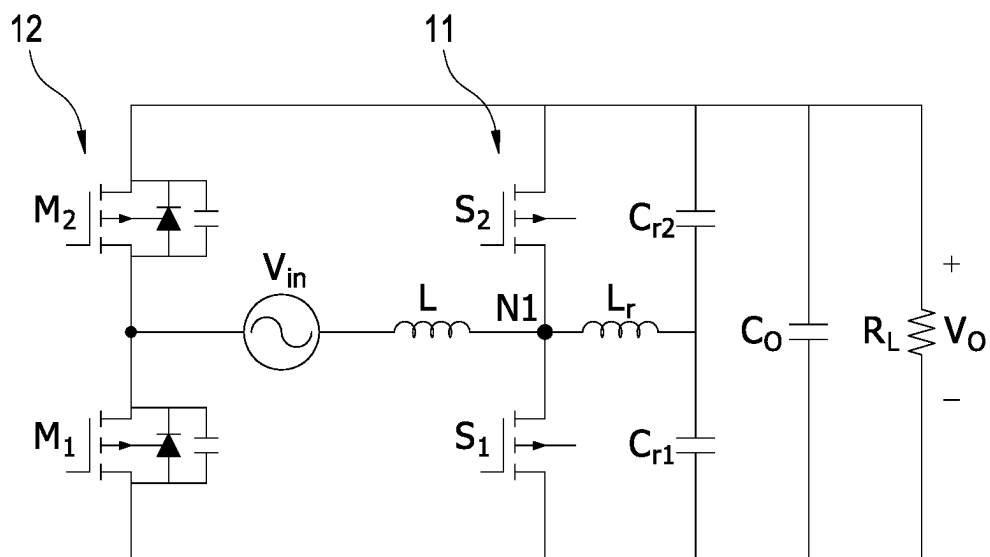


FIG.1C

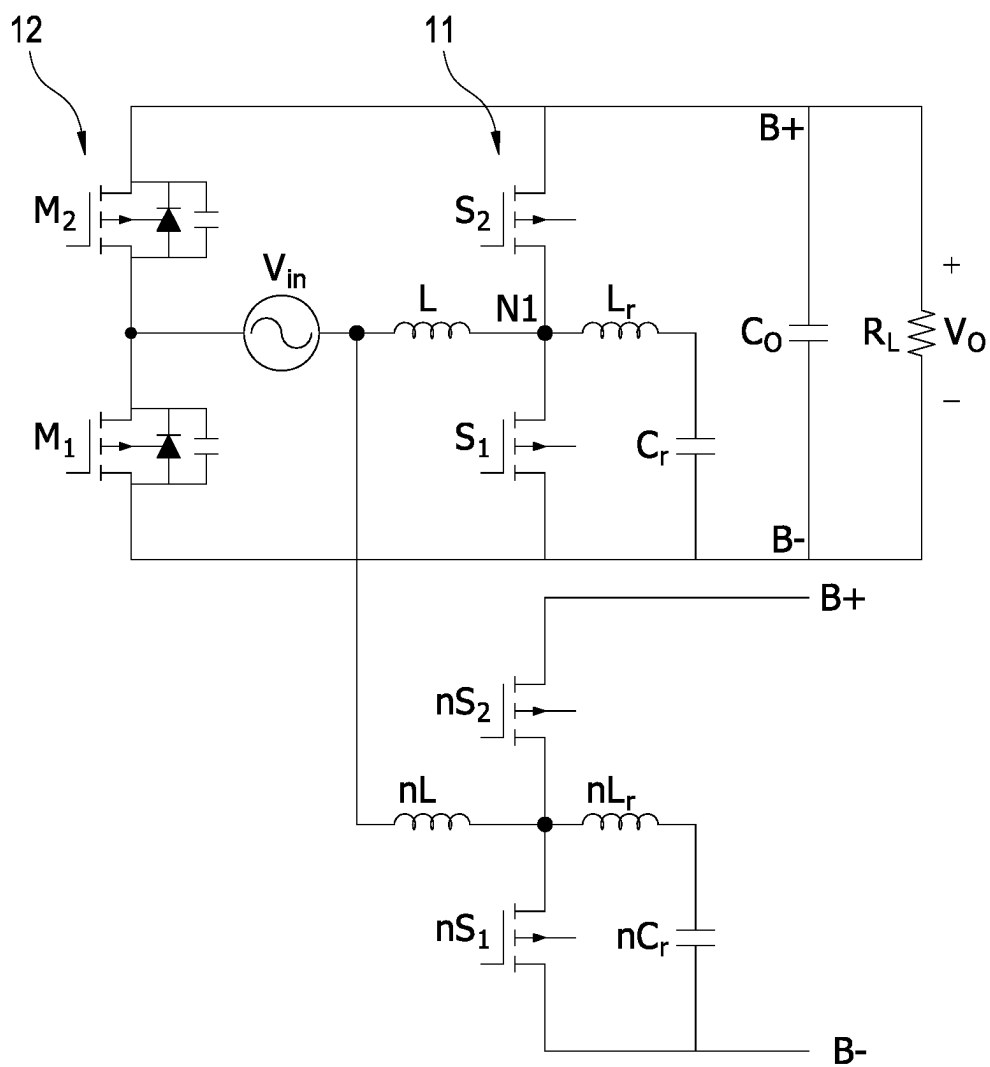


FIG. 2

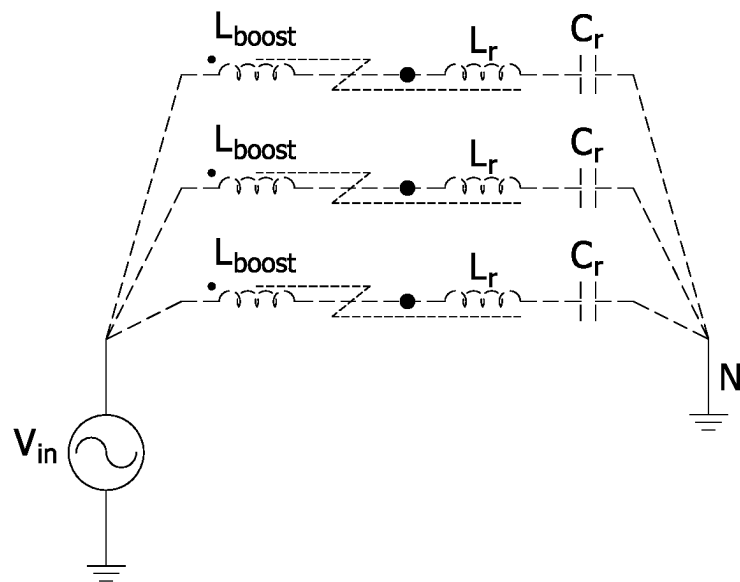


FIG.3A

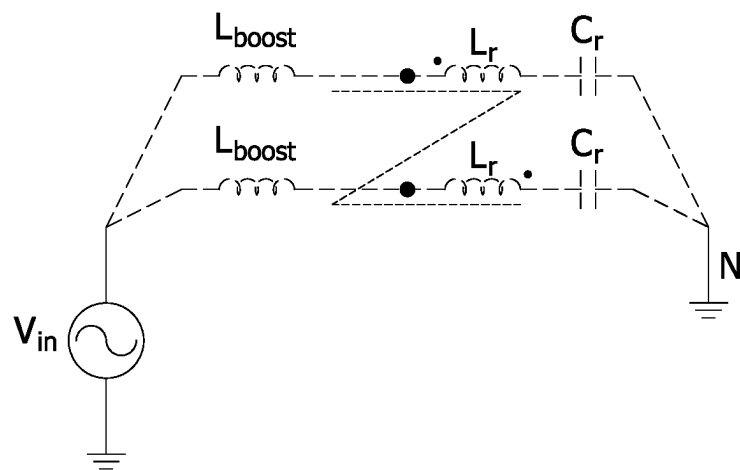


FIG.3B

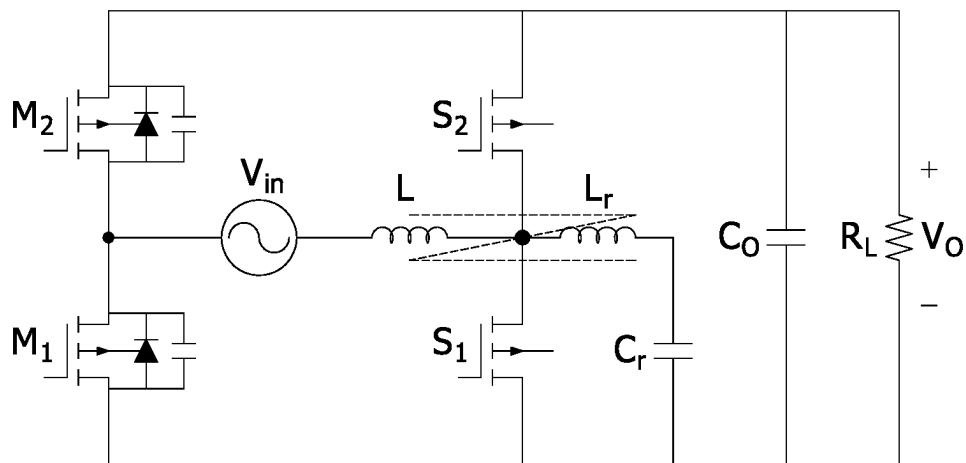


FIG. 4

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TOTEM-POLE POWER FACTOR CORRECTOR WITH ZERO-VOLTAGE SWITCHING

BACKGROUND

Technical Field

The present disclosure relates to a Totem-Pole power factor corrector, and more particularly to a Totem-Pole power factor corrector with zero-voltage switching.

Description of Related Art

The statements in this section merely provide background information related to the present disclosure and do not necessarily constitute prior art.

For a Totem-Pole power factor corrector (TPPFC), traditionally, the power density is increased by increasing the switching frequency. However, under the continuous conduction mode (CCM) in the conventional approach, it must be used to reduce heat generation since the heat generated by the hard switching of the switches increases with the switching frequency. However, due to the reduction of the switching frequency, although the generation of heat energy can be reduced, the effect of reducing the volume of the magnetic component cannot be achieved. Therefore, the operation mode of critical conduction mode (CRM) is proposed, which has the characteristics of zero-voltage switching (ZVS), thereby improving the operation frequency and increasing the efficiency. However, this operation mode (CRM) has following disadvantages:

1. The drastic frequency change makes the control more difficult, and the total input harmonic distortion (THD) is higher.

2. The high-frequency ripple and the low-frequency main power components exist at the same time so that it is not easy to design the filtering function of magnetic components.

3. Since the peak current in CRM operation is twice that of conventional CCM, the volume of the magnetic component needs to be larger.

4. Due to the large input ripple, if only a single-leg switch operates, a larger electromagnetic interference inductor (EMI choke) is required so the advantage of size will be lost. Therefore, two-phase interleaved operation is often used to solve this problem, but this increases the cost and control complexity, resulting in the problem of using multiple switches for low power.

Therefore, the present disclosure provides a Totem-Pole power factor corrector with zero-voltage switching to add at least one set of LC resonant tank to the circuit structure of the existing Totem-Pole power factor corrector (TPPFC), thereby solving the disadvantages of the existing technology under the continuous conduction mode (CCM) operation.

SUMMARY

An objective of the present disclosure is to provide a Totem-Pole power factor corrector with zero-voltage switching to solve the problems of existing technology.

In order to achieve the above-mentioned objective, the Totem-Pole power factor corrector with zero-voltage switching receives an input power source and converts the input power source into an output power source. The Totem-Pole power factor corrector includes an input inductor, a fast-switching switch leg, a slow-switching switch leg, a

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resonant tank, and an output capacitor. A first end of the input inductor receives the input power source. The fast-switching switch leg includes a fast-switching upper switch and a fast-switching lower switch, and the fast-switching upper switch and the fast-switching lower switch are commonly coupled at a first middle node. The slow-switching switch leg is coupled in parallel to the fast-switching switch leg, and the slow-switching switch leg includes a slow-switching upper switch and a slow-switching lower switch. The resonant tank includes a resonant inductor and at least one resonant capacitor. A first end of the resonant inductor is coupled to the first middle node, and a second end of the resonant inductor is coupled to the at least one resonant capacitor. The output capacitor is coupled in parallel to the fast-switching switch leg, and outputs the output power source.

In one embodiment, the number of the at least one resonant capacitor is one; a first end of the resonant capacitor is coupled to the second end of the resonant inductor, and a second end of the resonant capacitor is coupled to the fast-switching upper switch and the slow-switching upper switch.

In one embodiment, the number of the at least one resonant capacitor is one; a first end of the resonant capacitor is coupled to the second end of the resonant inductor, and a second end of the resonant capacitor is coupled to the fast-switching lower switch and the slow-switching lower switch.

In one embodiment, the number of the at least one resonant capacitor is two, which are a first resonant capacitor and a second resonant capacitor; a first end of the first resonant capacitor and a first end of the second resonant capacitor are coupled to the second end of the resonant inductor; a second end of the first resonant capacitor is coupled to the fast-switching lower switch and the slow-switching lower switch; a second end of the second resonant capacitor is coupled to the fast-switching upper switch and the slow-switching upper switch.

In one embodiment, a first end of the fast-switching upper switch and a first end of the fast-switching lower switch are commonly coupled to the first middle node; a first end of the slow-switching upper switch and a first end of the slow-switching lower switch are commonly coupled to the input power source; a second end of the fast-switching upper switch is coupled to a first end of the slow-switching upper switch; a second end of the fast-switching lower switch is coupled to a second end of the slow-switching lower switch.

In one embodiment, the input inductor and the resonant inductor form an integrated coupling structure.

In one embodiment, the Totem-Pole power factor corrector operates in a N-phase structure. The N-phase Totem-Pole power factor corrector includes N sets of the input inductors, N sets of the fast-switching switch legs, and N sets of the resonant tanks, and one set of the slow-switching switch leg and one set of the output capacitor.

In one embodiment, the input inductor in each phase is correspondingly coupled with the resonant inductor of the resonant tank.

In one embodiment, the two resonant inductors between two phases are cross-coupled.

Accordingly, the Totem-pole power factor corrector with zero-voltage switching of the present disclosure has following advantages: 1. The LC resonant tank is added to make the Totem-Pole power factor corrector have full switch zero-voltage switching function, thereby reducing switching loss to increase power conversion efficiency; 2. The LC resonant tank is added to accelerate the release of the electric

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energy stored in the parasitic capacitance of the switch, thereby suppressing the surge when the switch is switched to protect the switch; 3. A small input inductor current ripple is implemented so that a smaller EMI level may be selected and used; 4. By the resonance of the internal resonant tank, the parasitic capacitance of the switch to be turned on can be discharged during the dead time of the switch so that it can achieve zero voltage conduction; 5. The zero-voltage switching is implemented to reduce heat dissipation requirements for switch modules; 6. The zero-voltage switching increases the switching frequency and reduces the size of the magnetic components; 7. The integrated coupling structure with the resonant inductor and the input inductor is used to acquire the benefits of magnetic flux cancellation so as to increase efficiency while increasing integration and power density; 8. Under the uses of coupling inductors, no additional magnetic component and no additional cost are required.

It is to be understood that both the foregoing general description and the following detailed description are exemplary, and are intended to provide further explanation of the present disclosure as claimed. Other advantages and features of the present disclosure will be apparent from the following description, drawings, and claims.

BRIEF DESCRIPTION OF DRAWINGS

The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawing as follows:

FIG. 1A is a circuit diagram of a Totem-Pole power factor corrector with zero-voltage switching according to a first embodiment of the present disclosure.

FIG. 1B is a circuit diagram of the Totem-Pole power factor corrector with zero-voltage switching according to a second embodiment of the present disclosure.

FIG. 1C is a circuit diagram of the Totem-Pole power factor corrector with zero-voltage switching according to a third embodiment of the present disclosure.

FIG. 2 is a block circuit diagram of the Totem-Pole power factor corrector with zero-voltage switching operating in a multi-phase parallel structure according to the present disclosure.

FIG. 3A is a circuit diagram of an inductor coupling of the Totem-Pole power factor corrector with zero-voltage switching operating in the multi-phase parallel structure according to a first embodiment of the present disclosure.

FIG. 3B is a circuit diagram of the inductor coupling of the Totem-Pole power factor corrector with zero-voltage switching operating in the multi-phase parallel structure according to a second embodiment of the present disclosure.

FIG. 4 is a circuit diagram of the inductor coupling of the Totem-Pole power factor corrector with zero-voltage switching operating in a single-phase parallel structure according to the present disclosure.

DETAILED DESCRIPTION

Reference will now be made to the drawing figures to describe the present disclosure in detail. It will be understood that the drawing figures and exemplified embodiments of present disclosure are not limited to the details thereof.

Please refer to FIG. 1A, FIG. 1B, and FIG. 1C, which respectively show circuit diagrams of a Totem-Pole power factor corrector with zero-voltage switching according to three embodiments of the present disclosure. The Totem-

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Pole power factor corrector with zero-voltage switching (hereinafter referred to as Totem-Pole power factor corrector) receives an input power source V_{in} and converts the input power source V_{in} into an output power source V_o . In this embodiment, the input power source V_{in} is an AC power source, and the converted output power V_o is a DC power source.

The Totem-Pole power factor corrector includes an input inductor L , a fast-switching switch leg **11**, a slow-switching switch leg **12**, a resonant tank **13**, and an output capacitor C_o .

The input inductor L has a first end and a second end. The first end of the input inductor L receives the input power source V_{in} , i.e., the AC power source. The fast-switching switch leg **11** includes a fast-switching upper switch S_2 and a fast-switching lower switch S_1 , and the fast-switching upper switch S_2 and the fast-switching lower switch S_1 are commonly coupled at a first middle node $N1$. In this embodiment, the fast-switching switch leg **11** is coupled to the AC input power source V_{in} through the input inductor L . The slow-switching switch leg **12** is coupled in parallel to the fast-switching switch leg **11**. The slow-switching switch leg **12** includes a slow-switching upper switch M_2 and a slow-switching lower switch M_1 . In this embodiment, the slow-switching switch leg **12** is directly coupled to the AC input power source V_{in} , and is controlled by a frequency of the mains (i.e., the frequency of the AC input power source V_{in}).

A first end of the fast-switching upper switch S_2 and a first end of the fast-switching lower switch S_1 are commonly coupled at the first middle node $N1$. A first end of the slow-switching upper switch M_2 and a first end of the slow-switching lower switch M_1 are commonly coupled to the input power source V_{in} . A second end of the fast-switching upper switch S_2 is coupled to a second end of the slow-switching upper switch M_2 , and a second end of the fast-switching lower switch S_1 is coupled to a second end of the slow-switching lower switch M_1 . Therefore, a dual-level structure is formed through this circuit topology.

The resonant tank **13** includes a resonant inductor L_r and at least one resonant capacitor C_r , C_{r1} , C_{r2} . A first end of the resonant inductor L_r is coupled to the first middle node $N1$, and a second end of the resonant inductor L_r is coupled to the at least one resonant capacitor C_r , C_{r1} , C_{r2} . The output capacitor C_o is coupled in parallel to the fast-switching switch leg **11**, and outputs the output power source V_o .

In the first embodiment shown in FIG. 1A, the number of the at least one resonant capacitor is one. A first end of the resonant capacitor C_r is coupled to the second end of the resonant inductor L_r , and a second end of the resonant capacitor C_r is coupled to the fast-switching upper switch S_2 and the slow-switching upper switch M_2 .

In the second embodiment shown in FIG. 1B, the number of the at least one resonant capacitor is one. A first end of the resonant capacitor C_r is coupled to the second end of the resonant inductor L_r , and a second end of the resonant capacitor C_r is coupled to the fast-switching lower switch S_1 and the slow-switching lower switch M_1 .

In the third embodiment shown in FIG. 1C, the number of the at least one resonant capacitor is two, which are a first resonant capacitor C_{r1} and a second resonant capacitor C_{r2} . A first end of the first resonant capacitor C_{r1} and a first end of the second resonant capacitor C_{r2} are coupled to the second end of the resonant inductor L_r . A second end of the first resonant capacitor C_{r1} is coupled to the fast-switching lower switch S_1 and the slow-switching lower switch M_1 . A

second end of the second resonant capacitor C_{r2} is coupled to the fast-switching upper switch S_2 and the slow-switching upper switch M_2 .

In particular, in the third embodiment shown in FIG. 1C, since two resonant capacitors C_{r1} , C_{r2} are used, a cross voltage of each resonant capacitor C_{r1} , C_{r2} may be relatively low, which is about half of a cross voltage of a single resonant capacitor in FIG. 1A and FIG. 1B. Therefore, the selected capacitance value of each resonant capacitor C_{r1} , C_{r2} may be relatively small. In one embodiment, the capacitance value of each resonant capacitor C_{r1} , C_{r2} is the same and half of the original, and therefore the current stress of each resonant capacitor C_{r1} , C_{r2} is only half of that of the single resonant capacitor.

Moreover, in the circuit structure of the third embodiment, the two resonant capacitors C_{r1} , C_{r2} provide two current paths for the inductor current (that is, the current flowing through the resonant inductor L_r) so that the service life of each resonant capacitor C_{r1} , C_{r2} may be increased.

In summary, the present disclosure enables the fast-switching upper switch S_2 and the fast-switching lower switch S_1 of the fast-switching switch leg 11 to have zero-voltage switching by adding an LC resonant tank and a power inductor (that is, the input inductor L). It is characterized in that under the TPPFC (Totem-Pole power factor corrector) structure, the resonant inductor L_r is connected between the half-bridge switch and the input inductor L, and then is connected in series to the resonant capacitor C_r to a positive end or a negative end of the DC output capacitor C_o , or then is connected to the two resonant capacitors C_{r1} , C_{r2} to the positive end and the negative end of the DC output capacitor C_o .

Please refer to FIG. 2, which shows a block circuit diagram of the Totem-Pole power factor corrector with zero-voltage switching operating in a multi-phase parallel structure according to the present disclosure. Different from FIG. 1A to FIG. 1C, the Totem-Pole power factor corrector with zero-voltage switching of the present disclosure is applied to a single-phase structure, and the present disclosure may also realize the multi-phase parallel structure shown in FIG. 2. Specifically, as shown in FIG. 2, when the Totem-Pole power factor corrector operates in a N-phase structure, the N-phase Totem-Pole power factor corrector includes N sets of the input inductors L, N sets of the fast-switching switch legs 11, and N sets of the resonant tanks 13, and one set of the slow-switching switch leg 12 and one set of the output capacitor C_o . As shown in FIG. 2, nL represents the Nth set of input inductor L, nS_1 represents the fast-switching lower switch S_1 of the Nth set of fast-switching switch leg 11, nS_2 represents the fast-switching upper switch S_2 of the Nth set of fast-switching switch leg 11, nL_r represents the resonant inductor L_r of the Nth set of resonant tank 13, and nC_r represents the resonant capacitor C_r of the Nth set of resonant tank 13. In particular, the output of the Nth set of circuit structure shown in FIG. 2 is coupled to both ends of the output capacitor C_o (respectively indicated by B+ and B-). Therefore, under the multi-phase parallel structure, the interleaved phase control can be used to make the ripple components of the input current be reduced because of counteracting the ripple components to each other.

Under the multi-phase parallel structure shown in FIG. 2, the inductor coupling has different embodiments. Please refer to FIG. 3A, which shows a circuit diagram of an inductor coupling of the Totem-Pole power factor corrector with zero-voltage switching operating in the multi-phase parallel structure according to a first embodiment of the

present disclosure. Specifically, the input inductor L boost in each phase is correspondingly coupled with the resonant inductor L_r of the resonant tank 13. Alternatively, please refer to FIG. 3B, which shows a circuit diagram of the inductor coupling of the Totem-Pole power factor corrector with zero-voltage switching operating in the multi-phase parallel structure according to a second embodiment of the present disclosure. Specifically, two resonant inductors L_r between two phases are cross-coupled.

Please refer to FIG. 4, which shows a circuit diagram of the inductor coupling of the Totem-Pole power factor corrector with zero-voltage switching operating in a single-phase parallel structure according to the present disclosure. In the present disclosure, the input inductor L and the resonant inductor L_r are two different inductors or two sets of coils are wound on two different iron cores. Alternatively, the input inductor L and the resonant inductor L_r form an integrated coupling structure, that is, the input inductor L and the resonant inductor L_r share the same iron core, and achieve an integrated structure of the common coupling magnetic path as shown in FIG. 4.

Therefore, the Totem-Pole power factor corrector with zero-voltage switching is used to add at least one set of LC resonant tank to the circuit structure of the existing Totem-Pole power factor corrector (TPPFC), thereby solving the disadvantages of the existing technology under the continuous conduction mode (CCM) operation.

In summary, the present disclosure has the following features and advantages:

1. The LC resonant tank is added to make the Totem-Pole power factor corrector have full switch zero-voltage switching function, thereby reducing switching loss to increase power conversion efficiency.
2. The LC resonant tank is added to accelerate the release of the electric energy stored in the parasitic capacitance of the switch, thereby suppressing the surge when the switch is switched to protect the switch.
3. A small input inductor current ripple is implemented so that a smaller EMI level may be selected and used.
4. By the resonance of the internal resonant tank, the parasitic capacitance of the switch to be turned on can be discharged during the dead time of the switch so that it can achieve zero voltage conduction.
5. The zero-voltage switching is implemented to reduce heat dissipation requirements for switch modules.
6. The zero-voltage switching increases the switching frequency and reduces the size of the magnetic components.
7. The integrated coupling structure with the resonant inductor and the input inductor is used to acquire the benefits of magnetic flux cancellation so as to increase efficiency while increasing integration and power density.
8. Under the uses of coupling inductors, no additional magnetic component and no additional cost are required.

Although the present disclosure has been described with reference to the preferred embodiment thereof, it will be understood that the present disclosure is not limited to the details thereof. Various substitutions and modifications have been suggested in the foregoing description, and others will occur to those of ordinary skill in the art. Therefore, all such substitutions and modifications are intended to be embraced within the scope of the present disclosure as defined in the appended claims.

What is claimed is:

1. A Totem-Pole power factor corrector with zero-voltage switching, configured to receive an input power source and convert the input power source into an output power source, the Totem-Pole power factor corrector comprising:

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an input inductor, a first end of the input inductor configured to receive the input power source,
 a fast-switching switch leg, comprising a fast-switching upper switch and a fast-switching lower switch, and the fast-switching upper switch and the fast-switching lower switch commonly coupled at a first middle node,
 a slow-switching switch leg, coupled in parallel to the fast-switching switch leg, and the slow-switching switch leg comprising a slow-switching upper switch and a slow-switching lower switch,
 a resonant tank, comprising a resonant inductor and at least one resonant capacitor; a first end of the resonant inductor coupled to the first middle node, and a second end of the resonant inductor coupled to the at least one resonant capacitor, and
 an output capacitor, coupled in parallel to the fast-switching switch leg, and configured to output the output power source.

2. The Totem-Pole power factor corrector as claimed in claim 1, wherein the number of the at least one resonant capacitor is one; a first end of the resonant capacitor is coupled to the second end of the resonant inductor, and a second end of the resonant capacitor is coupled to the fast-switching upper switch and the slow-switching upper switch.

3. The Totem-Pole power factor corrector as claimed in claim 1, wherein the number of the at least one resonant capacitor is one; a first end of the resonant capacitor is coupled to the second end of the resonant inductor, and a second end of the resonant capacitor is coupled to the fast-switching lower switch and the slow-switching lower switch.

4. The Totem-Pole power factor corrector as claimed in claim 1, wherein the number of the at least one resonant capacitor is two, which are a first resonant capacitor and a second resonant capacitor; a first end of the first resonant capacitor and a first end of the second resonant capacitor are

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coupled to the second end of the resonant inductor; a second end of the first resonant capacitor is coupled to the fast-switching lower switch and the slow-switching lower switch; a second end of the second resonant capacitor is coupled to the fast-switching upper switch and the slow-switching upper switch.

5. The Totem-Pole power factor corrector as claimed in claim 1, wherein a first end of the fast-switching upper switch and a first end of the fast-switching lower switch are commonly coupled to the first middle node; a first end of the slow-switching upper switch and a first end of the slow-switching lower switch are commonly coupled to the input power source; a second end of the fast-switching upper switch is coupled to a first end of the slow-switching upper switch; a second end of the fast-switching lower switch is coupled to a second end of the slow-switching lower switch.

6. The Totem-Pole power factor corrector as claimed in claim 1, wherein the input inductor and the resonant inductor form an integrated coupling structure.

7. The Totem-Pole power factor corrector as claimed in claim 1, wherein the Totem-Pole power factor corrector operates in a N-phase structure;

the N-phase Totem-Pole power factor corrector comprises:

N sets of the input inductors, N sets of the fast-switching switch legs, and N sets of the resonant tanks, and one set of the slow-switching switch leg and one set of the output capacitor.

8. The Totem-Pole power factor corrector as claimed in claim 7, wherein the input inductor in each phase is correspondingly coupled with the resonant inductor of the resonant tank.

9. The Totem-Pole power factor corrector as claimed in claim 7, wherein the two resonant inductors between two phases are cross-coupled.

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